LOS/NLOS Classification and Distance Estimation for UWB Indoor Positioning Using Feature-Engineered ML and Hybrid CNN-Transformer

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***Abstract*—Accurate line-of-sight (LOS) and non-line-of-sight (NLOS) identification is essential for precise ultra-wideband (UWB) indoor positioning. We study two-path LOS/NLOS classification and path-wise distance estimation on the Decawave DWM1000 dataset. After preprocessing, a valid second path is detected in 40,864 of 41,996 usable samples (97.3%), producing an expanded two-path dataset of 83,992 rows. On the feature-engineered pipeline, XGBoost gives the best single-model AUC (0.9832), while stacked ensembling attains the highest accuracy (0.9413) with marginal AUC change. A separate raw-CIR CNN+Transformer pipeline reaches AUC 0.9827 on the original split, improving to 0.9862 with CIR augmentation, but this deep-learning result is not directly comparable to the ML results because it uses a different pipeline and split. For regression, path 1 is moderately predictable (best RMSE 1.3706 m, $R^2 = 0.6698$), whereas path 2 attains near-perfect scores because its target is derived from input features supplied to the model. These results support tree-based supervised models for this dataset, while unsupervised clustering remains weak.**

***Index Terms*—UWB, LOS/NLOS classification, channel impulse response, CNN, Transformer, indoor positioning, distance estimation**

I. Introduction

Ultra-wideband (UWB) technology has emerged as a leading solution for centimeter-level indoor positioning due to its high time resolution and resistance to multipath fading [1]. However, non-line-of-sight (NLOS) propagation conditions—caused by obstacles such as walls, furniture, and human bodies—introduce systematic ranging errors that severely degrade positioning accuracy [2]. Identifying whether a received signal has traveled via a line-of-sight (LOS) or NLOS path is therefore a critical prerequisite for accurate indoor localization.

A. Problem Statement

The core challenge in UWB indoor positioning is that NLOS propagation adds a positive bias to the measured range, because signals travel longer paths via reflection, diffraction, or scattering. Without reliable LOS/NLOS classification, ranging errors of several meters can persist, rendering downstream position estimates unreliable [3]. The problem is compounded in dense indoor environments where multiple propagation paths coexist in the channel impulse response (CIR), and the dominant path may not be the direct one.

B. State-of-the-Art Solutions

Existing approaches to LOS/NLOS classification fall into two broad categories.

Feature-engineered methods. Traditional approaches extract hand-crafted features from the CIR—such as rise time, kurtosis, energy ratios, and peak-to-noise metrics—and feed them to classifiers like support vector machines (SVMs), k-nearest neighbors (KNN), or random forests [3], [4]. These methods are well-studied and interpretable, but rely heavily on domain expertise for feature selection and may miss subtle signal patterns.

Deep learning methods. Recent work applies convolutional neural networks (CNNs) directly to the raw CIR waveform, bypassing manual feature engineering [5]. However, standard CNNs capture only local patterns. Transformer architectures [6], which model long-range dependencies through self-attention, have recently been applied to UWB NLOS detection [10], [11] and show promise for capturing relationships between widely separated multipath components.

C. Limitations of Existing Approaches

Despite their effectiveness, current methods exhibit several weaknesses: (1) feature-engineered approaches treat each CIR measurement as a single-path decision, discarding the rich information in secondary propagation paths; (2) CNN-only deep learning models lack the ability to relate distant CIR components, limiting their capacity to capture multipath relationships; and (3) most studies evaluate on the original balanced dataset without examining how classification extends to multi-path scenarios with realistic class imbalance.

D. Proposed Solution

In this paper, we address these limitations through a dual-pipeline approach:

TABLE I
DATASET FEATURES

Feature	Description
RANGE	Measured distance (m); also regression target
FP_IDX	First path index in CIR
FP_AMP1–3	First path amplitude estimators
STDEV_NOISE	Standard deviation of noise
CIR_PWR	Total CIR power
MAX_NOISE	Maximum noise level
RXPACC	Preamble accumulation count
FRAME_LEN	Frame length
PREAM_LEN	Preamble length
CIR0–CIR1015	1016-sample channel impulse response

- A **two-path extraction and labeling framework** that extends binary LOS/NLOS classification to the two dominant signal paths in each CIR measurement, producing an expanded dataset with realistic class imbalance.
- A **feature engineering pipeline** producing 25 per-path features for classical ML models (Logistic Regression, SVM, Random Forest, Gradient Boosted Trees, XGBoost, and ensembles), with RFECV used to examine feature-subset efficiency.
- A **hybrid 1D-CNN + Transformer architecture** that operates directly on the raw 1016-sample CIR waveform, combining local feature extraction with global self-attention in a separate end-to-end pipeline.
- **Distance estimation models** for both dominant paths, interpreted with attention to the analytic constraint in the path 2 target definition.

II. PROPOSED SOLUTION DESCRIPTION

A. Dataset and Data Preparation

We use the publicly available UWB LOS/NLOS dataset [7] collected using Decawave DWM1000 modules across seven distinct indoor environments. The dataset contains 42,000 samples (21,000 LOS and 21,000 NLOS), each comprising 14 scalar features and a 1016-sample CIR recorded at 1 ns resolution. Table I summarizes the key features.

Our preprocessing pipeline performs the following steps:

CIR normalization. Each CIR sample is divided by its corresponding RXPACC value to normalize the preamble accumulation effect.

Constant/metadata removal. Three raw features (CH, BITRATE, PRFR) are constant and removed. We additionally drop loader metadata (ENV_ID) to avoid using environment identifiers as predictors, leaving the 11 scalar features used downstream.

Feature scaling. The 11 remaining scalar features are standardized using zero-mean unit-variance scaling (StandardScaler).

NaN/Inf handling. Any NaN or Inf values introduced during CIR normalization are replaced with zero.

Train/test split. After removing 4 degenerate samples, the usable set has 41,996 rows and is split stratified 80/20 into 33,596 training and 8,400 test samples.

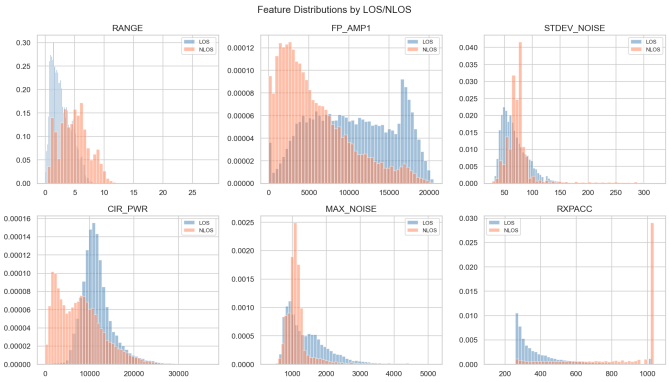

Fig. 1. Distribution of scalar features by LOS (blue) and NLOS (orange) class. Features such as CIR_PWR and FP_AMP1 show strong class separability.

Algorithm 1 Two-Path Peak Detection

Require: CIR waveform $c[0..1015]$, first path index FP_IDX
Ensure: Path indices p_1, p_2 and amplitudes a_1, a_2

1: $p_1 \leftarrow \arg\max_{j \in [\text{FP_IDX}-10, \text{FP_IDX}+10]} c[j]$
2: $a_1 \leftarrow c[p_1]$
3: Smooth c with a 3-sample moving average to obtain $\tilde{c}$
4: Find peaks in $\tilde{c}$ with height $> 0.02 \cdot a_1$, distance > 15, prominence $> 0.02 \cdot a_1$
5: Remove peaks within ± 15 samples of p_1
6: $p_2 \leftarrow$ remaining peak with maximum prominence (fallback: max amplitude)
7: $a_2 \leftarrow c[p_2]$ (from original unsmoothed CIR)
8: **return** p_1, a_1, p_2, a_2

Fig. 1 shows the distribution of key features by LOS/NLOS class, revealing clear separability for several features. Correlation analysis shows that FP_AMP1–3 are highly correlated ($r > 0.9$), as are STDEV_NOISE and MAX_NOISE ($r \approx 0.95$).

B. Two-Path Extraction and Labeling

UWB CIR measurements often contain multiple propagation paths. The two shortest (dominant) paths carry the most information for positioning: if the first path is LOS, the second is necessarily NLOS (reflected); if the first path is NLOS, both paths are NLOS. Algorithm 1 describes our peak detection procedure.

After preprocessing, 41,996 samples remain usable for two-path extraction, and a second path is detected in 40,864 of them (97.3%). Fig. 2 shows example CIR waveforms with detected peaks.

Two-path labeling. For each usable original sample, we generate two rows: path 1 retains the original LOS/NLOS label, while path 2 is always labeled NLOS: if path 1 is LOS, then path 2 is necessarily a reflected component, and if path 1 is itself NLOS, then path 2 is also NLOS. This yields an expanded dataset of 83,992 rows with a 25%/75% LOS/NLOS class imbalance.

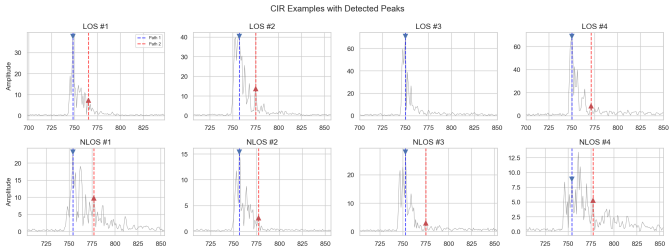

Fig. 2. Example CIR waveforms showing detected first path (triangle) and second path (circle) for LOS and NLOS conditions.

TABLE II
KEY LOCAL PER-PATH ENGINEERED FEATURES (8 OF THE 13 PER-PATH DESCRIPTORS)

Feature	Description
path_idx	Peak sample index in CIR
path_amp	Peak amplitude
rise_time	Samples from 10% threshold to peak
decay_time	Samples from peak to 10% threshold
kurtosis_local	Kurtosis of ± 15-sample window
energy_ratio	Local energy / total CIR energy
peak_to_noise	Peak amplitude / STDEV_NOISE
amplitude_ratio	This path amp / other path amp

C. Feature Engineering Pipeline

For each detected path, we compute 8 local waveform features (Table II) together with 5 full-CIR statistical descriptors, 9 shared scalar features, and 3 derived shared features, yielding a 25-dimensional feature vector per path.

Feature importance analysis indicates that amplitude- and noise-related descriptors dominate. RFECV over the Random Forest estimator returns the full 25-feature set as optimal under the 3-fold cross-validated accuracy criterion, i.e. no strict feature reduction is supported by the data; however, the RFECV curve is nearly flat from roughly ten features onward, so a smaller subset would sacrifice little accuracy in practice. Tree-based importance rankings still highlight noise-normalized peak strength as the main discriminator, and SHAP analysis [16] (Fig. 4) confirms these rankings from a model-agnostic perspective.

The classical ML study includes Logistic Regression (LR), Support Vector Machine (SVM), Random Forest (RF) [9], Gradient Boosted Trees (GBT), and XGBoost (XGB) [8], followed by simple averaging and stacked ensembles over the strongest tree-based models.

D. End-to-End Deep Learning Pipeline

We propose a hybrid 1D-CNN + Transformer architecture that operates directly on the raw 1016-sample CIR waveform. Algorithm 2 describes the training procedure.

Architecture. The model consists of three components:

1) CNN encoder: A multi-scale first convolutional stage (parallel kernels 3/7/15) is followed by MaxPool(2), halving the sequence from 1016 to 508. Two subsequent residual Conv1D blocks [14] (kernels 5 and 3) with max-pooling

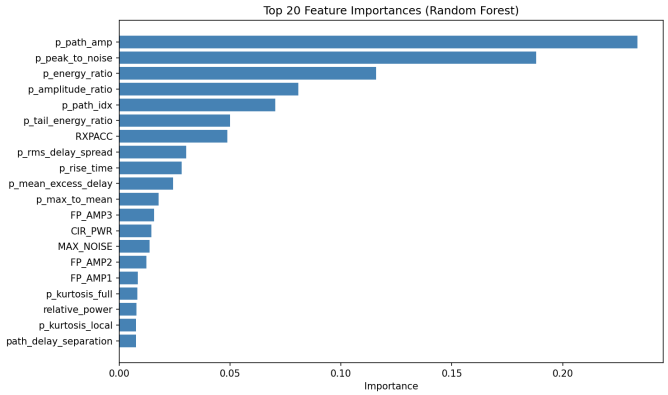

Fig. 3. Top 15 features ranked by Random Forest Gini importance. Signal power and noise-related features dominate.

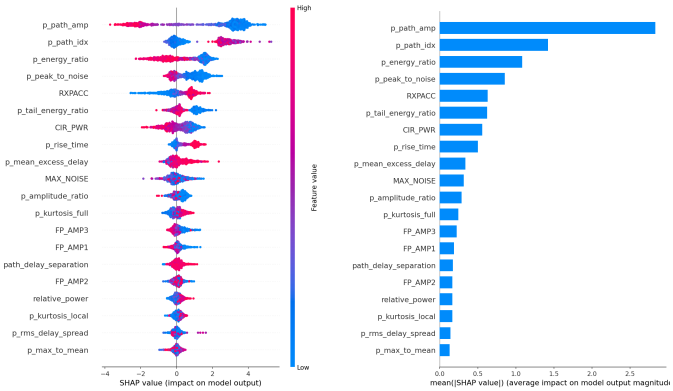

Fig. 4. SHAP feature impact: beeswarm plot (left) showing per-sample feature contributions and mean absolute SHAP values (right) confirming Gini-based rankings.

progressively reduce this to 127 feature vectors of dimension 128.

2) Transformer encoder: Two Transformer encoder layers with 4-head self-attention model global dependencies across the CNN-encoded sequence. Sinusoidal positional encoding preserves temporal ordering. The self-attention mechanism allows the model to relate distant CIR components, capturing multipath relationships that span the full waveform.

3) Classification head: Global average pooling reduces the transformer output to a single 128-dimensional vector, which is concatenated with 11 scaled scalar features and passed through a two-layer MLP (hidden size 64) to produce a binary logit.

The DL model is trained on the original balanced 42,000-sample dataset (not the two-path expanded set), using an 80/20 stratified split in a separate pre-computed raw-CIR pipeline.

III. EXPERIMENTAL RESULTS AND DISCUSSION

A. Classification Results

Table IV summarizes the supervised classification results. The feature-engineered ML models and ensembles are evaluated on the two-path expanded test set, while the CNN+Transformer results come from a separate raw-CIR

TABLE III
CNN-TRANSFORMER HYPERPARAMETERS

Parameter	Value
CNN channels	$32 \to 64 \to 128$
CNN kernels	3/7/15 (parallel), then 5 and 3
Transformer layers	2
Attention heads	4
Feedforward dim	256
MLP hidden	64
Dropout	0.1
Optimizer	Adam [15] (lr = 1×10^{-3})
Loss	BCEWithLogitsLoss
Label smoothing	$\varepsilon = 0.05$ ($y' = 0.975/0.025$) [12]
Batch size	256
Early stopping	patience = 5
Device	Auto-select (CUDA/MPS/CPU)

Algorithm 2 CNN-Transformer Training

Require: Training set $\{(\mathbf{c}_i, \mathbf{s}_i, y_i)\}_{i=1}^{N}$ (CIR, scalars, label)
Ensure: Trained model f_θ

```
Initialize f_θ with random weights
best_loss ← ∞; patience_count ← 0
for epoch = 1 to 50 do
    for each mini-batch B do
        y'_B ← y_B · 0.95 + 0.025 {label smoothing, ε=0.05}
        ŷ ← f_θ(c_B, s_B)
        L ← BCEWithLogits(ŷ, y'_B)
        Update θ via Adam with ∇_θ L
    end for
    Compute validation loss L_val
    if L_val < best_loss then
        best_loss ← L_val; save θ; reset patience
    else
        patience_count ← patience_count + 1
    end if
    if patience_count ≥ 5 then
        break (early stopping)
    end if
end for
return f_θ with best validation weights
```

pipeline on the original split. The latter should therefore be read as a related reference point rather than a direct one-to-one comparison.

Among directly comparable feature-engineered models, XGBoost is best by AUC (0.9832), while stacked ensembling gives the highest accuracy (0.9413) with essentially no AUC gain over XGBoost alone. SVM, once its inputs are standardized inside an sklearn pipeline, lands close to the Logistic Regression baseline but still below the tree ensembles, consistent with kernel methods benefiting from nonlinearity yet plateauing on this already feature-engineered representation. In the separate raw-CIR pipeline, CIR augmentation yields a measurable AUC improvement for the CNN+Transformer while SMOTE [13] has essentially no effect on the ML models; however, this DL-vs-ML comparison must remain qualified because the underlying splits and inputs differ. Fig. 5 and Fig. 6 show representative ROC and confusion-matrix views for the main baseline models.

TABLE IV
CLASSIFICATION PERFORMANCE

Model	Accuracy	AUC	Test Set
Logistic Regression	0.9185	0.9730	Expanded
SVM	0.9262	0.9735	Expanded
Random Forest	0.9396	0.9821	Expanded
Gradient Boosted Trees	0.9312	0.9821	Expanded
XGBoost	0.9398	0.9832	Expanded
Ensemble Avg	0.9399	0.9832	Expanded
Ensemble Stacked	0.9413	0.9829	Expanded
CNN+Transformer	0.9368	0.9827	Original split
CNN+Transformer + CIR Aug	0.9387	0.9862	Original split

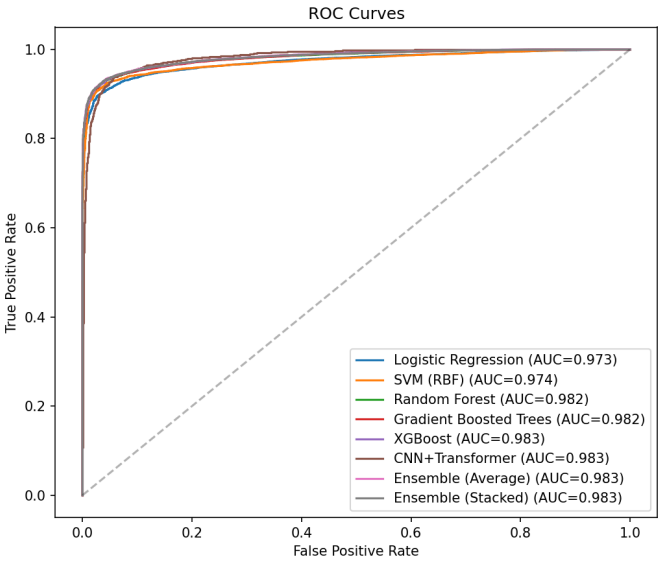

Fig. 5. ROC curves for selected baseline classification models. Tree-based ML and CNN+Transformer achieve similarly high AUC values.

B. Distance Estimation Results

Four regressors are trained per path: Ridge Regression (linear baseline), RF, GBT, and XGB. Table V presents their performance. Path 1 uses the original test split, while path 2 uses the 8,192 valid path 2 test samples retained after preprocessing and two-path extraction.

Path 1 regression is materially harder than classification because the target RANGE includes environment-dependent NLOS bias [3], [4], with the best results coming from RF/XGB-style tree ensembles at roughly 1.37 m RMSE and $R^2 \approx 0.67$. Path 2 appears much easier, but this must be interpreted carefully: the target is defined as $\texttt{RANGE} + (\texttt{path2_idx} - \texttt{FP_IDX}) \times 0.2998$ m/ns (speed of light), while RANGE is also supplied as an input. The near-perfect path 2 scores therefore reflect a strong analytical constraint in the target construction rather than an entirely independent prediction task. Fig. 7 shows predicted vs. actual distance scatter plots.

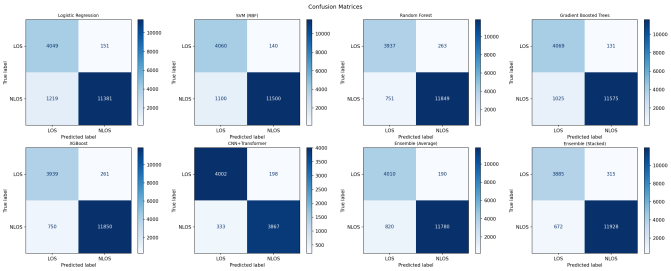

Fig. 6. Confusion matrices for selected baseline classification models.

TABLE V
DISTANCE ESTIMATION PERFORMANCE

Path	Model	RMSE (m)	R^2
Path 1	Ridge	1.6164	0.5407
	Random Forest	1.3706	0.6698
	GBT	1.3877	0.6615
	XGBoost	1.3715	0.6693
Path 2	Ridge*	0.0235	1.0000
	Random Forest	0.4658	0.9997
	GBT	0.2900	0.9999
	XGBoost	1.2238	0.9980

*Perfect score: target is a near-deterministic linear function of inputs.

C. ML vs. Deep Learning Comparison

The principal supervised finding is that feature-engineered tree models perform well on the expanded two-path task: XGBoost reaches AUC 0.9832, and stacked ensembling yields only marginal improvement. The CNN+Transformer also performs strongly, reaching AUC 0.9827 on the original split and 0.9862 with CIR augmentation, but this ML-vs-DL comparison is not direct because the deep model is trained and evaluated in a separate raw-CIR pipeline on the original split. Fig. 9 summarizes the model comparison.

Accordingly, we treat the DL results as complementary evidence that raw-CIR learning is viable, rather than as evidence that DL surpasses ML on an identical benchmark. Within the notebook experiments, SMOTE has negligible effect on the ML pipeline; CIR augmentation in the deep-learning pipeline produces a similarly small accuracy shift but a clearer AUC gain, indicating that waveform-level augmentation primarily sharpens the model's probability ranking rather than its top-1 decisions.

D. Interpretability via Attention

A key advantage of the CNN-Transformer is interpretability through attention weight visualization. Fig. 10 shows a representative attention pattern for sample CIR waveforms. Qualitatively, elevated attention often appears near first-arrival and multipath regions, offering suggestive evidence about which CIR segments influence predictions. This complements the feature-importance analysis of the classical models.

E. Error Analysis

Classification errors primarily occur in ambiguous cases where NLOS signals have strong direct-path-like characteristics or where LOS signals experience significant multipath interference. The gap between LR and the tree-based models indicates a nonlinear decision boundary; once its features are standardized inside an sklearn pipeline, the RBF-kernel SVM recovers most of that gap and lands close to LR but still below the tree ensembles, consistent with kernel methods benefiting from nonlinearity yet plateauing on this already feature-engineered representation.

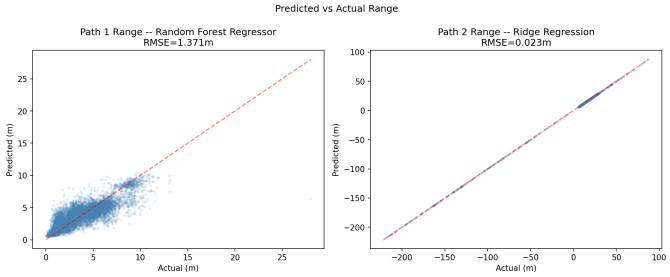

Fig. 7. Predicted vs. actual distance for path 1 (left) and path 2 (right) regression models.

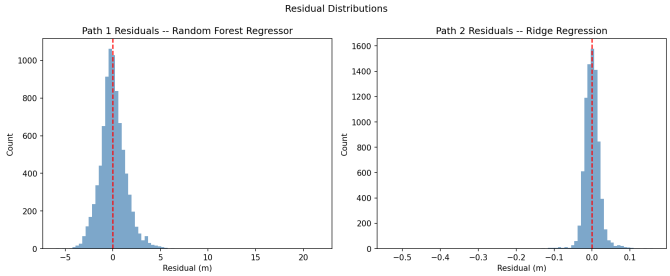

Fig. 8. Regression residual distributions. Path 1 shows a slight positive bias from NLOS ranging error; path 2 residuals are tightly clustered around zero, consistent with the analytically constrained target.

For distance estimation, path 1 remains the more meaningful regression task because the measured range includes environment-dependent bias that is not trivially recoverable from a single formula. By contrast, path 2 regression benefits from the direct geometric relationship embedded in the target definition and reflects the analytical dependency rather than genuine predictive difficulty.

Unsupervised clustering is weak. KMeans reaches only 0.5700 test accuracy with ARI -0.0099 (essentially random agreement), and DBSCAN fragments the scaled feature space into 9 small density clusters that do not align with LOS/NLOS labels ($\sim$12.7% of points rejected as noise and 0.5263 accuracy on the clustered points). These results suggest that LOS/NLOS separation in this feature space is not readily achievable through unsupervised methods alone.

From a practical standpoint, the CNN-Transformer operates on the full 1016-sample CIR plus 11 scalar features, avoiding manual feature engineering, whereas the tabular models require the 25-dimensional engineered feature vector.

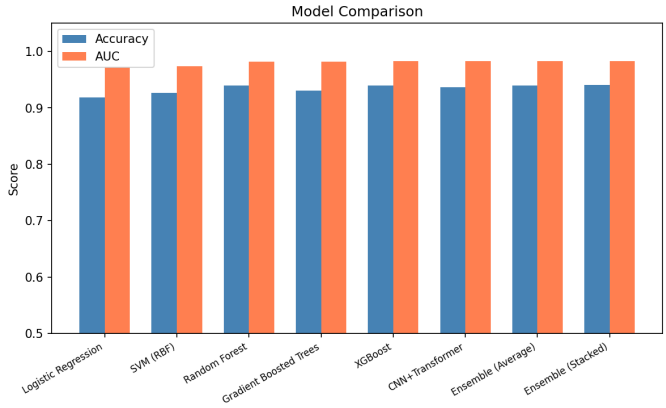

Fig. 9. Comparison of classification metrics across all models.

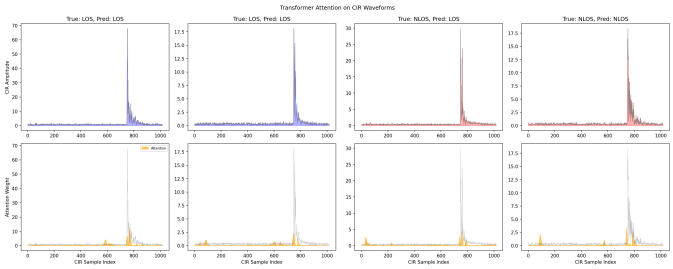

Fig. 10. Transformer attention weights overlaid on a CIR waveform. High attention concentrates near the first path and multipath components.

IV. Conclusion

We presented a concise UWB LOS/NLOS classification and distance-estimation study using a feature-engineered two-path pipeline together with a separate raw-CIR CNN+Transformer reference pipeline. The two-path extraction stage finds a valid second path in 97.3% of usable preprocessed samples and expands the supervised dataset to 83,992 rows.

The key findings are as follows: (1) among directly comparable ML models, XGBoost gives the best single-model AUC (0.9832), while stacked ensembling gives the highest accuracy (0.9413) with marginal AUC change; (2) the CNN+Transformer is competitive, and CIR augmentation improves it further, but this DL result comes from a separate original-split raw-CIR pipeline and is not a strict one-to-one comparison with the ML results; (3) path 1 regression is moderately accurate, whereas path 2 regression is analytically constrained by the way its target is defined; and (4) clustering results are weak and should not be used to claim strong unsupervised separability.

A. Future Work

Several directions merit further investigation: (1) Ensemble methods combining ML and DL predictions; (2) Cross-environment generalization studies using leave-one-environment-out validation; (3) Model compression and quantization for real-time deployment on embedded UWB receivers; (4) Extension to multi-class scenarios distinguishing between different NLOS propagation mechanisms (diffraction, reflection, scattering).

Acknowledgment

The authors thank the contributors of the UWB-LOS-NLOS-Data-Set for making the dataset publicly available.

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